

Short Answer test

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Ans

Computer Vision is a branch of Artificial intelligence (AI) that enables computers to capture, process, analyze and understand the image or videos like humans. It uses image processing, pattern recognition and machine learning algorithm to identify objects, recognize faces, detect movements, and make decision based on visual data.

a) Main objective of computer vision

- To acquire and process digital images.
- To recognize and classify objects.
- To understand scenes and extract useful information.
- To automate visual inspection and information.

Two real-world application:

- Face recognition in Smartphones and Security systems.

Self-driving (Autonomous) Vehicles.

2. Computer Vision extracts meaningful information from images and videos using advanced algorithms. It helps computers identify objects, patterns, shapes, colors and movements accurately.

a) Feature extraction: Feature extraction is the process of identifying important characteristics such as edges, corners, textures and shapes from an image for further analysis.

b) Object recognition: Object recognition identifies and classifies objects present in an image. It improves computer vision by enabling accurate detection and decision making.

Image Processing image enhancement

consists of 3 levels

- Low level processing: image enhancement, noise removal, brightness adjustment.
 - Mid level processing: image enhancement, edge detection, feature extraction.
 - High level processing: defect recognition, object understanding, image interpretation.
- 5 levels understanding: It improves image quality by removing noise and enhancing brightness and contrast.
- Mid level
Extracts features from image
- High level
Understands image meaning
- Polymers objects recognition and scene analysis
- Perform segmentation and edge detection

4. A digital image is made up of picture elements called pixels. Each pixel stores brightness or color information. The quality of an image depends on its resolution.

a) Pixel: A pixel is the smallest unit of a digital image.

b) image resolution: Image resolution is the no. of pixels in an image. Higher resolution gives better image quality.

Image sensing and acquisition is the process of capturing a real world scene using camera and converting it into a digital image.

a) Image sensor: A image sensor converts light into electrical signals.

- b) Two image acquisition devices
- Digital camera.
 - Scanner.

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Sampling converts a continuous image into pixels, and quantization assigns intensity value to each pixel. Together they create a digital image.

a) Sampling: Sampling is the process of dividing an image into pixels.

b) Quantization: Quantization assigns brightness or color values to each pixel.

The image information model explains how light passes through a camera lens and forms an image on the sensor.

a) Perspective projection: It converts a 3D scene into a 2D image making distant objects appear smaller.

b) Illumination: Illumination provides light which is necessary to capture a clear image.

8. Image processing improves image quality while Computer Vision helps computer understand image.

a) Image processing is the process of enhancing digital image.

b) Difference

• Image Processing $\rightarrow$ improves image
• Computer Vision $\rightarrow$ understands image

9. Computer Vision is used in many fields such as healthcare, security etc.

Healthcare: Detects diseases from X-ray

Autonomous: Detects roads, traffic sign

10. Digital image fundamentals include pixel sampling and resolution, which help computer analyze image.

a) Pixel: The smallest unit of a digital

Higher resolution gives clearer image
Sampling divides the image into n